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EDUCATION

Aptech, Advanced Diploma in Software Engineering (ADSE)

January 2024 – June 2026

- **Relevant Coursework:** Data Structures & Algorithms, Object Oriented Programming, Machine Learning, Data Science, Fullstack Web & App Development, Database Systems, System Design, Cloud Computing, Critical & Analytical thinking.

EXPERIENCE

HNG Tech, Software Engineer Intern

October 2025 – December 2025

- Architected a distributed notification microservice (Redis Streams pub/sub, idempotent delivery), spearheading a 3-engineer pod, reducing notification latency by **43%** (p95 <300ms).
- Contributed to Legal WatchDog's AI compliance engine using graph traversal to map regulatory dependencies, reducing **manual compliance** efforts by **57%** and accelerating audit cycles by **3×**.
- Integrated two A2A (Agent-to-Agent) compliant AI agents (a riddle agent and a real-time news update agent) into [Telex.im](https://telex.im).

Snapilabs, Software Developer Intern

August 2024 – November 2024

- Built SupaNote's AI workspace landing page with virtualized scrolling and debounced search, achieving **98+** Lighthouse score and **40%** lower bounce rate.
- Rebuilt Krea AI's UI from static mockups using component composition, delivering pixel perfect responsive implementation 2 weeks ahead of schedule, accelerating feature development velocity by **33%**.

PROJECTS

Odyssey, Decoder-Only Transformer Research (LLaMA-style)

July 2026

Python, PyTorch, Rust, YAML, pytest, mpy, Matplotlib

- Built decoder-only transformer components from first principles: owned byte-level BPE tokenizer, embeddings (32K×768), RoPE, RMSNorm, SwiGLU, and GQA attention (12/4 heads) under frozen Spec v1.0.0 for Odyssey-Tiny (**768-dim, 12 layers, 2K context**).
- Validated numerical parity with Phalanx Runtime (Rust) via automated cross-impl tests for RoPE, RMSNorm, SwiGLU, and GQA (**max error $\leq 1.3 \times 10^{-4}$; mean $\approx 2.9 \times 10^{-6}$**) across 8 experiments, aligning PyTorch training with Rust inference.
- Shipped **125+ tests** with reproducible configs and CPU benchmarks (**~38K tok/s attention, ~22K tok/s SwiGLU**); documented tensor shapes, GQA tradeoffs, and paper-backed architecture notes.

Finomaly, Real-time Financial Anomaly Detection System

June 2026

Python, PyTorch, PyTorch Geometric, Kafka, Redis, FastAPI, Docker

- Engineered real-time fraud detection pipeline processing 6.3M+ transactions using a dual-model architecture: a PyTorch Autoencoder (ROC-AUC 0.951) and a GraphSAGE GNN for structural anomaly detection, achieving **99.76% recall** and **4.8ms** per-transaction scoring **latency**.
- Built streaming inference layer with Redpanda (Kafka API) and Redis-cached GNN embeddings, delivering end-to-end detection in **<10ms** at **p95** and publishing flagged transactions to an alert topic consumed by downstream systems.
- Developed FastAPI REST API with **model explainability**, attributing each alert to the AE or GNN branch with human-readable reasons (balance inconsistencies, fraud-ring proximity) and validated with an ablation study proving both branches contribute complementary signal

Rails, Virtual Account Reconciliation Infrastructure on Nomba

July 2026

Typescript, Fastify, BullMQ, NombaAPI, PostgreSQL, Redis, Render

- Engineered a 5-state payment reconciliation engine (matched, underpaid, overpaid, misdirected, duplicate) on Nomba virtual accounts, with HMAC-SHA256 webhook verification, **idempotent deduplication** on transactionId/sessionId, and sub-account-scoped VA provisioning; validated in production with a live transfer reconciled to matched in under **3 seconds** end-to-end.
- Built a dual-process Fastify API + BullMQ worker that acknowledges Nomba webhooks with HTTP 200 immediately, queues reconciliation asynchronously, and retries failed jobs with **5-attempt exponential backoff**; tuned BullMQ worker polling to **cut idle Redis usage ~85%** on Upstash; delivered signed outbound transfer.* events to downstream subscribers with persistent delivery audit logs.
- Developed an 8-endpoint multi-tenant REST API with SHA-256 hashed API key auth, OpenAPI/Swagger (Bearer auth), and customer ledger APIs (paginated transactions + filterable statements), enabling Nomba VA reconciliation integration in **<1 hour** without rebuilding webhook/edge-case handling.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, Java, Go, Dart, C++, Rust, Swift

Frameworks & Technologies: FastAPI, Django, Spring Boot, NestJS, Next.js, React, Flutter, Gin

Databases & Platforms: PostgreSQL, MongoDB, MySQL, Supabase, Firebase

Cloud/DevOps: Docker, Render, Railway, Vercel, Git

Tools: Postman, Figma, Expo, Android Studio, XCode